Architecture and Workflows

flowchart TD

UI[Streamlit Engineering Workbench] --> API[FastAPI REST]

API --> C[PipelineCoordinator]

C --> PDF[PyMuPDF PDF extraction]

PDF --> CLEAN[Text cleaning]

CLEAN --> CHUNK[Hierarchical chunking]

CHUNK --> EM[Sentence-Transformer embeddings]

EM --> DB[(ChromaDB vector index)]

CHUNK --> ENT[SWC/Port/Interface rule extraction]

ENT --> SQL[(SQLite structured entities)]

C --> RET[Semantic Retriever top-k]

RET --> DB

RET --> PROMPT[Grounded Prompt Builder]

PROMPT --> LLM[MockLLM or Gemini]

LLM --> VERIFY[Citation ID Verifier]

VERIFY --> API

SQL --> API

Ingestion: PDF → page/block text → cleaning → chunks → embeddings → vector index + deterministic entity extraction → SQLite.

Q&A: question → query embedding → ChromaDB nearest neighbors → grounded prompt → provider → cited answer → Python citation validation → answer/evidence in Streamlit.

Source of truth: backend/coordinator.py (../backend/coordinator.py), core/rag (../core/rag), frontend/app.py (../frontend/app.py).

The citation check validates the *reference ID*, not semantic entailment. Human reviewers must open the cited source before accepting engineering claims.